MIL-T-82378C(TD)
7 March 1983
SUPERSEDING
MIL-T-82378B
6 May 1981

MILITARY SPECIFICATION

TRAINING AID, RIFLE, M16A1, TWO TO ONE SCALE
DEVICE 3F68B

This specification is approved for use by the Naval Training Equipment Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the construction of a two times enlarged mock-up of the M16A1 Rifle, Device 3F68B.

1.2 For Army procurement Device 3F68B carries the designator Device 09-19. The subject specification applies to both device numbers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. Unless otherwise noted, the effective dates of issue of the applicable documents shall be those listed in the Department of Defense Index of Specifications and Standards (DoDISS) of 1 July 1982 and Supplement of 1 November 1982.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-794	Parts and Equipment; Procedure for Packaging and Packing of

DRAWINGS

Naval Training Equipment Center

Ben Lorenz Assoc., DL 8D-128-600	Drawings, Mock-up M16A1 Rifle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Training Equipment Center, Orlando, FL 32813, Attn: Code N212, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document, or by letter.

FSC 6910-N416

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

TM 9-10005-249-10	Operator's Manual, M16A1 Rifle
UL 94	Standard Tests for Flammability (Underwriter's Laboratories, Inc) 207 East Ohio Street Chicago, IL 60611

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer (PCO).)

3. REQUIREMENTS

3.1 Materials.- Materials used in the fabrication of the device shall be in accordance with Drawings on DL: 8D-128-600. Materials not specified on the drawings shall be as specified herein.

3.1.1 Metals.- Metals shall be either corrosion-resistant or suitably treated to resist corrosion under atmospheric conditions specified in 3.3.

3.1.1.1 Stainless steel shall be type 303 or 416.

3.1.1.2 Aluminum sheet shall be 6061-T6.

3.1.2 Wood.- Wood shall be selected grades of hardwood lumber, suitable for paint finishes. Wood shall be properly seasoned and free from checks, splits, and warp.

3.1.2.1 Plywood.- Plywood shall have a minimum of five plies or solid core plywood, both surfaces to be of hardwood; all plies to be free of spaces or checks and joined by waterproof glue.

3.1.2.2 Glue.- Glue used shall not be adversely affected by moisture, lacquer or paint.

3.1.3 Hardboard.- Hardboard shall be tempered, smooth on both sides and moisture resistant, for a paint or lacquer-finish. U.S. Gypsum "Duron" or material equivalent in strength, rigidity and smoothness of surface is acceptable.

3.1.4 Plastic.- Plastics shall be durable and resistant to cracking, brittleness, discoloration and deformation upon aging, or when subject to the range of temperatures specified in 3.3. Flammability shall comply with Underwriter's Laboratory UL94.

3.1.4.1 Cements for plastics.- Cements used for plastics shall be those recommended by the manufacturer of the plastic and tested by the contractor for the use as required by 3.3. The cemented joints shall form a strong, homogeneous bond.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

3.2 Design.- Device 3F68B shall be designed for use on table top as shown in Figure 1. The Device 3F68B consists of all the major components listed under 3.6. This includes a kit of replacement and repair parts.

3.3 Reliability.- The device shall withstand continuous classroom use and shall not be adversely affected in any manner during storage and transportation when under ambient conditions of temperature ranging from -65°F to +166°F, and of relative humidity ranging up to 95 percent. The device shall be so constructed that no components, assemblies, or subassemblies shall bind or become undesirably free. No degradation shall be caused in the mechanical performance stipulated for the device because of shipment, handling, or storage.

3.4 Maintainability.- The device shall be so constructed that the device will be durable and require no adjustments beyond the adjustments normally made in the operational M16A1 rifle. Components subject to fatigue, breakage, or wear shall be replaced out of the kit for repair parts. The replacement kit stock shall permit use of the device for 15 years. No special tools are required to maintain the device.

3.5 Performance.- The device, being a scale model of the operational rifle, shall allow the demonstration of all operational modes, field stripping, and maintenance of the M16A1 rifle. See also Operator's Manual TM 9-10005-249-10.

3.6 Details of components.- The device shall consist of the following major components:

(a) Rifle M16A1, mock-up

(b) Ammunition rounds, simulated

(c) Shipping - storage case

(d) Kit of repair parts.

3.6.1 Rifle M16A1 mock-up.- The rifle mock-up shall be an operating, partly sectionalized, two times actual scale model of the M16A1 rifle. The rifle mock-up shall be constructed in conformance to the drawings listed under 2.1.

3.6.1.1 A magazine with a capacity of twenty simulated rounds shall be supplied.

3.6.1.2 A bipod as shown on the drawings under 2.1 shall be supplied.

3.6.1.3 The components for the table top support, shown in Figure 1, consisting of stock support, wood strap, and four C-clamps shall be supplied.

3.6.1.4 A plastic container for catching ejected rounds shall be supplied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

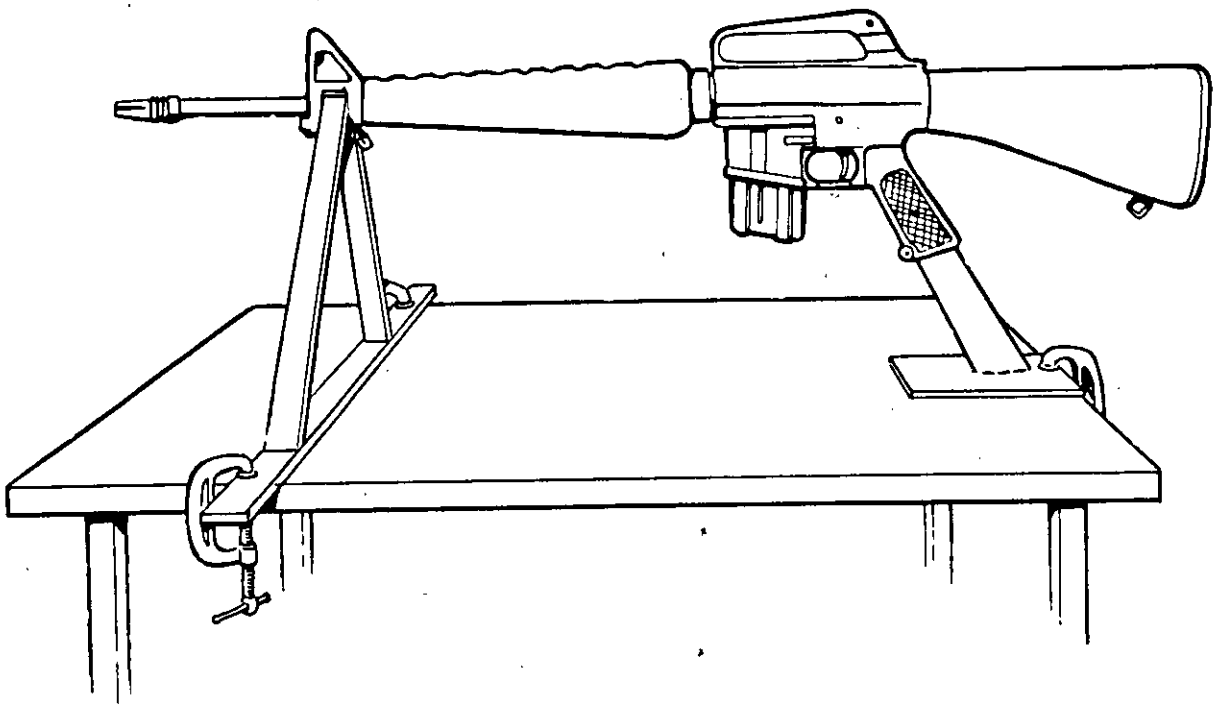


Figure 1. Device 3F68B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

3.6.1.5 A deflector made from clear plastic for directing ejected cartridges shall be be supplied.

3.6.2 Ammunition rounds.- Ammunition rounds shall conform to the drawings under 2.1. A total of 30 shall be supplied as part of the rifle mock-up.

3.6.3 Shipping-storage case.- The storage case shall conform to the drawings under 2.1. The case shall accommodate all components of the device, including the spare parts supplied with the device. Components shall be secured in the case without the use of tools. Marking labels shall be provided indicating the location of each component. The case shall have suitable and adequate draw-type fasteners and carrying handles.

3.6.3.1 Case.- The case shall be made from grey high-density polyethelene. Ikelheimer-Ernst "Fiberbilt" case, Style 852 or equal is acceptable. Overall dimensions shall be L: 60 inches, W: 9 1/2 inches, H: 14 3/4 inches.

3.6.3.2 Markings.- On the top outer surface a marking plate shall be fastened carrying the following legend in black, block letters:

DEVICE 09-19(3F68B)
M16A1 RIFLE MOCK-UP
SCALE 2:1

3.6.4 Kit for repair parts.- The kit shall be packed in a polyethelene bag in the shipping case. The kit contains the following parts:

Name	Quantity	Drawing No.
Spring, plunger	1	8D-128-140-8
Spring, dust cover	1	8D-128-140-3
Spring, sear	1	8D-128-140-4
Spring, trigger	1	8D-128-140-6
Spring, hammer	1	8D-128-140-7
Spring, magazine	1	8D-128-160-5
Spring, disconnect	2	8D-128-460-11
Cotter pin, bolt	2	Commercial.

3.7 Finish.- All surfaces to be painted shall be sealed and primed. All wood surfaces shall be finished with a satin-surface lacquer. Metal surfaces to be painted shall be finished with a satin epoxy or baked-on enamel. Application of finishes shall be accomplished in such a manner as not to impair the performance of the equipment.

3.7.1 All finished surfaces shall be free from tool marks, burrs, sharp edges, or visible imperfections.

3.8 Color.- Colors used shall be as noted in the drawings of 2.1 and be such as to easily distinguish one component from another, to indicate and trace systems, and to accent the principles presented.

3.9 Nameplates or product markings.-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

3.9.1 Nameplate and emblem.- A Government-furnished nameplate and two emblems shall be affixed to the device in accordance with the drawings of 2.1.

3.9.2 Markings.- All markings for settings that appear on the operational rifle M16A1 shall be effected by engraving or by recessed molding in the mock-up. The markings shall be filled in with color for contrast with the background and shall be clearly legible.

3.10 Workmanship.- The device, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, riveting, machine-screw assemblage, welding, brazing, marking of parts and assemblies, plating, sanding, grinding; painting; and freedom from sharp edges or burrs, or any other damage or defect that could make the part or equipment unsatisfactory for the operation or function intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Facilities.- The contractor shall furnish any facilities, equipment or personnel that the Government may require to insure that the training aid meets the requirements of this specification.

4.3 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.4 In-process inspection.- In-process inspection shall include such visual and mechanical examination, and testing of materials, subassemblies, parts and accessories (including purchased standard items) during the manufacturing process, as may be necessary to assure conformance to all the requirements of this specification.

4.5 Quality conformance inspection.- Quality conformance inspection shall consist of examinations and tests of the First Article and the production units as outlined under 4.5.1 and 4.5.2.

4.5.1 First Article inspection.-

4.5.1.1 Examinations.- The training aid shall be examined for conformance to the following requirements of this specification:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

(a) Materials

(b) Design

(c) Details of components. The examination shall verify that individual components correspond to the drawings supplied or drawings properly updated and approved in accordance to contract requirements

(d) Finish and color

(e) Nameplates and product markings

(f) Workmanship.

4.5.1.2 Tests.- The following tests shall be performed to check compliance with requirements under 3.5:

(a) Remove device from shipping storage case and set-up for operation on table

(b) Operate rifle with two magazine loads of 16 rounds each through a full 32 cycles of feeding, firing, extracting, and ejecting. One cycle with automatic, one with semi-automatic setting. If failure occurs in any part of this test, the whole test shall be repeated after malfunction is removed

(c) Field strip device and reassemble in accordance with TM 9-1005-249-10.

4.5.2 Production inspection.-

4.5.2.1 Examination.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government accepted First Article.

4.5.2.2 Tests.- Tests, as outlined under 4.5.1.2, shall be conducted on each production unit.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Packaging requirements are referred to under 3.6.3. The requirements of MIL-STD-794 shall apply.

5.2 Packing.- Units shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

cushioning, blocking, and bracing shall be provided within shipping containers during shipment, handling, and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The device is intended for use in classroom training of military personnel in the operation, adjustment, assembly, and disassembly of the Rifle M16A1.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

CUSTODIAN:
Navy (TD)

PREPARING ACTIVITY:
Navy (TD)
DOD Project No. 6910-N416

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDING OFFICER
NAVAL TRAINING EQUIPMENT CENTER
ATTN: CODE N-212
ORLANDO, FL. 32813

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:04Z
Check the source to verify that this is the current version before use.